

Section 3.8 Master Algorithm for LPG Semantic Communication: A Coupled Adaptive Loop Architecture

3.8.1 Overview and Architectural Rationale

This section presents the Master Algorithm for Logarithmic Probability Geometry (LPG) Semantic Communication — a complete, formally specified, and reproducible algorithmic pipeline that integrates every theoretical structure developed in this framework. The algorithm is designed to serve two simultaneous purposes: (i) as a theoretical instrument proving that the LPG framework is computationally coherent and internally consistent, and (ii) as a simulation blueprint enabling any independent investigator to reproduce the results claimed herein using standard engineering tools including NVIDIA Sionna, PyTorch, and the DeepSC baseline.

The algorithm adopts Architecture B: a Coupled Adaptive Loop, in which the sender agent and receiver agent are not independent processing modules but co-evolving systems connected by a bidirectional feedback channel. This architectural choice is motivated by three structural commitments of the LPG framework: the dynamic nature of semantic permeability μ_s , the context-dependence of the interpretation map ι , and the requirement that semantic equilibrium be reached through iterative convergence rather than single-shot transmission.

The S-OSI protocol stack is represented throughout as a single semantic encoding/decoding transform pair, encapsulating all seven S-OSI layers inside two operators. This design decision preserves modularity while keeping the master algorithm's control flow transparent and directly simulatable.

3.8.2 Structural Inventory and Layer Dependency

The pipeline integrates eight structural layers in strict dependency order. Table 3.8.1 specifies each layer, its theoretical structure, and its functional role in the algorithm.

Layer	Structure	Formal Object	Role in Pipeline
1	sem quadruple	$(\mathbf{B}, \mathbf{W}, \Gamma, \iota)$	Atomic unit of transmission
2	Semantic manifold	$\mathcal{M}_s$	Geometric space of all possible meanings
3	Complex meaning map	$\mathbf{M}(\mathbf{m}, \mathbf{c}, \text{sem}) \in \mathbb{C}$	Encodes meaning as complex number
4	Complex analysis	Cauchy–Riemann conditions; $H^2(\mathbb{D})$	Analyticity gate and regularity constraint

Layer	Structure	Formal Object	Role in Pipeline
5	Conservation laws	$\mathcal{E}_{\text{sem}}(\mathbf{t})$	Semantic energy invariant across rounds
6	Field theory	$\mu\mathbf{s}, \varepsilon\mathbf{s}, \mathbf{H}\mathbf{s}$	Propagation dynamics and channel model
7	S-OSI stack	$\mathcal{E}\text{SOSI}, \mathcal{D}\text{SOSI}$	Layered encoding and decoding transform
8	Adaptive agents	$\pi\mathbf{s}(\mathbf{t}), \pi\mathbf{r}(\mathbf{t}), \theta\mathbf{r}(\mathbf{t})$	Policy learning and feedback adaptation

Table 3.8.1: Structural inventory of the LPG Semantic Communication pipeline.

3.8.3 Preliminary Definitions

The following objects are globally defined before the algorithm executes and are shared between both agents unless otherwise stated.

Shared vocabulary: $W = \{w_1, w_2, \dots, w_{|W|}\}$ is known to both sender and receiver. This constitutes the shared semantic codebook.

The sender state at round t is the triple:

$$\mathbf{s}_t = (\Gamma_t, \pi\mathbf{s}(t), \theta\mathbf{s}(t))$$

where $\Gamma_t \subseteq W$ is the active anchor set, $\pi\mathbf{s}(t)$ is the anchor selection policy, and $\theta\mathbf{s}(t)$ are the sender's learnable parameters.

The receiver state at round t is the quadruple:

$$\mathbf{r}_t = (\hat{\Gamma}_t, \hat{\iota}_t, \pi\mathbf{r}(t), \theta\mathbf{r}(t))$$

where $\hat{\Gamma}_t$ is the reconstructed anchor set, $\hat{\iota}_t$ is the reconstructed interpretation map, $\pi\mathbf{r}(t)$ is the reconstruction policy, and $\theta\mathbf{r}(t)$ are the receiver's learnable parameters.

The feedback signal at round t is defined as:

$$\delta t = \mathcal{F}_s(t) - \mathcal{F}^*$$

where $\mathcal{F}^* \in (0, 1)$ is the target semantic fidelity threshold set prior to simulation. The coupled loop drives $\delta t \rightarrow 0$.

3.8.4 The S-OSI Encoding and Decoding Transforms

The entire S-OSI protocol stack is encapsulated as two operators. The encoding transform maps a sem instance to a complex-valued transmitted signal:

$$\mathcal{E}\text{SOSI} : \text{sem} \rightarrow \psi \quad (\text{semantic encoding})$$

$$\mathcal{D}\text{SOSI} : \tilde{\psi} \rightarrow \text{sem} \quad (\text{semantic decoding})$$

The encoding operator is realized through the complex meaning map:

$$M(m, c, \text{sem } t) = \lambda^T u(m, Bt) + i (\lambda(c) - \lambda)^T u(m, Bt)$$

where $u(m, Bt) \in \mathbb{R}|W|$ is the log-probability vector induced by message m under base space Bt , $\lambda \in \mathbb{R}|W|$ is the global semantic weight vector, and $\lambda(c) \in \mathbb{R}|W|$ is the context-modulated weight vector. The full transmitted complex signal vector is $\psi t = \mathcal{E}\text{SOSI}(\text{sem}t) \in \mathbb{C}d$, where d is the transmission dimension.

A critical constraint governs the encoding step. The complex signal ψt is transmitted if and only if the Cauchy–Riemann semantic regularity conditions are satisfied:

$$\partial \text{Re}(M)/\partial B_j = \partial \text{Im}(M)/\partial \lambda_j(c), \quad \partial \text{Re}(M)/\partial \lambda_j(c) = -\partial \text{Im}(M)/\partial B_j$$

for all $j = 1, \dots, |W|$. These conditions constitute the analyticity gate of the pipeline. Transmissions violating these conditions are either corrected or the round is aborted. This constraint has no analogue in any existing semantic communication system.

3.8.5 Semantic Field Propagation Model

The channel is not modelled as a simple additive noise pipe. In the LPG framework the channel is a semantic field governed by the propagation equation:

$$\nabla^2 \psi - \mu_s \epsilon_s (\partial^2 \psi / \partial t^2) = -J_s$$

where μ_s is semantic permeability, ϵ_s is semantic permittivity, and J_s is the semantic source current density encoding the sender's intent. The received signal is:

$$\tilde{\psi}t = H_s \psi t + \eta t, \quad \eta t \sim \mathcal{CN}(0, \sigma^2 I)$$

where $H_s \in \mathbb{C}d \times d$ is the semantic channel matrix. This matrix decomposes as:

$$H_s = \mu_s \cdot H_{\text{phys}} \odot H_{\text{sem}}$$

where H_{phys} is the physical fading matrix, H_{sem} encodes the semantic alignment between sender and receiver knowledge states, and $\odot$ denotes the elementwise (Hadamard) product. This decomposition makes explicit that semantic communication is governed by two independent channel effects: physical propagation and semantic alignment.

3.8.6 Semantic Conservation Law

The LPG framework imposes a conservation law at the semantic layer with no counterpart in classical information theory. Define the semantic energy at round t as:

$$\mathcal{E}_{\text{sem}}(t) = \|\psi t\|^2 = \sum_j |M_j(m, c, \text{sem}t)|^2$$

The conservation condition requires that between consecutive transmission rounds the semantic energy drift does not exceed a permitted bound $\Delta \mathcal{E} > 0$:

$$|\mathcal{E}_{\text{sem}}(t) - \mathcal{E}_{\text{sem}}(t-1)| \leq \Delta \mathcal{E}$$

Transmissions violating this condition are flagged as conservation anomalies and excluded from the capacity estimator. This enforcement is a direct algorithmic consequence of the field-theoretic structure of the LPG framework.

3.8.7 Fidelity and Capacity Measurement

Semantic fidelity at round t is measured via the Jensen–Shannon divergence between the sender’s true interpretation map ι_t and the receiver’s reconstructed map $\hat{\iota}_t$:

$$\mathcal{F}_s(t) = 1 - D_{JS}(\iota_t \parallel \hat{\iota}_t)$$

where $D_{JS} \in [0, 1]$ is the Jensen–Shannon divergence computed in bits. Perfect semantic fidelity corresponds to $\mathcal{F}_s(t) = 1$. The empirical semantic capacity after T rounds is:

$$\hat{C}_s(T) = \log_2(|W| / (|W| - \mu[\Gamma](T))) \cdot \hat{\mu}_s(T)$$

where the empirical anchor density and semantic permeability estimates are:

$$\mu[\Gamma](T) = (1/T) \sum_t |\Gamma_t|, \quad \hat{\mu}_s(T) = \bar{\mathcal{F}}_s(T) \cdot \mu[\Gamma](T) / |W|$$

3.8.8 Adaptive Policy Updates

The distinguishing feature of Architecture B is that both agents and the field parameter μ_s update after every transmission round. Three coupled update rules govern adaptation.

Sender policy update (multiplicative weights):

$$\pi_s(t+1)(\Gamma) \propto \pi_s(t)(\Gamma) \cdot \exp(\eta_s \cdot \delta t \cdot p[\Gamma_t = \Gamma])$$

The sender reinforces anchor sets that produced positive feedback ($\delta t > 0$) and suppresses those producing negative feedback ($\delta t < 0$).

Receiver parameter update (gradient ascent on fidelity):

$$\theta_r(t+1) = \theta_r(t) + \eta_r \cdot \nabla_{\{\theta_r\}} \mathcal{F}_s(t)$$

Semantic permeability update (field parameter adaptation):

$$\mu_s(t+1) = \mu_s(t) + \gamma \cdot \delta t$$

This third update rule closes the field-theoretic loop: fidelity error updates not just the agents but the physical field parameter itself. Semantic permeability is therefore not a fixed channel constant but a learned quantity that converges to the channel’s true semantic permeability at equilibrium.

3.8.9 Convergence Criterion and System Output

The coupled adaptive loop terminates when the feedback signal stabilizes within a convergence tolerance $\epsilon_{conv} > 0$ for τ_{conv} consecutive rounds:

$$|\delta t| < \epsilon_{conv} \text{ for } \tau_{conv} \text{ consecutive rounds}$$

At convergence the system outputs the complete semantic channel characterization:

$$OUTPUT = (\hat{C}_s(T), \bar{\mathcal{F}}_s(T), \mu_s(T), \Gamma^*, \iota^*)$$

This output tuple constitutes the full empirical characterization of the LPG semantic channel: its capacity, fidelity performance, permeability, the converged anchor set, and the converged interpretation map.

3.8.10 Master Algorithm: Complete Formal Specification

Algorithm 3.1 presents the complete master algorithm. Every step is formally specified and directly maps onto a simulatable computational procedure.

Algorithm 3.1: LPG Semantic Communication — Coupled Adaptive Loop

Require: $W, \mathcal{P}^*, \eta_s, \eta_r, \gamma, \varepsilon_{\text{conv}}, \tau_{\text{conv}}, T_{\text{max}}$

Ensure: $\hat{C}_s, \hat{\mathcal{F}}_s, \mu_s^*, \Gamma^*, \hat{r}^*$

INITIALIZATION

$t \leftarrow 1; \mu_s(1) \leftarrow \mu_s, 0; \pi_s(1) \leftarrow \text{Uniform}(2W); \theta_r(1) \leftarrow \theta_r, 0$

WHILE $t \leq T_{\text{max}}$ AND NOT converged DO

— SENDER —

- S1. Sample anchor set: $\Gamma_t \sim \pi_s(t)$
- S2. Construct base signal: $B_t \leftarrow B(\Gamma_t, W)$
- S3. Form sem instance: $\text{semt} \leftarrow (B_t, W, \Gamma_t, t)$
- S4. Verify Cauchy–Riemann: ABORT round if violated
- S5. Encode via S-OSI: $\psi_t \leftarrow \mathcal{E}\text{SOSI}(\text{semt})$
- S6. Check conservation: FLAG anomaly if $|\mathcal{E}(t) - \mathcal{E}(t-1)| > \Delta \mathcal{E}$
- S7. Transmit ψ_t over H_s

— CHANNEL —

- C1. Propagate: $\tilde{\psi}_t \leftarrow H_s \psi_t + \eta_t, \eta_t \sim \text{CN}(0, \sigma^2 I)$

— RECEIVER —

- R1. Invert signal: $\hat{B}_t \leftarrow \text{Re}(H_s^\dagger \tilde{\psi}_t)$
- R2. Reconstruct Γ : $\Gamma_t \leftarrow \text{ML-GreedySearch}(\hat{B}_t, W, \sigma^2)$
- R3. Recover $\hat{r}$: $\hat{r}_t \leftarrow \text{Softmax}(B(\Gamma_t, W))$
- R4. Measure fidelity: $\mathcal{F}_s(t) \leftarrow 1 - D_{\text{JS}}(r_t \parallel \hat{r}_t)$
- R5. Compute feedback: $\delta_t \leftarrow \mathcal{F}_s(t) - \mathcal{P}^*$
- R6. Send δ_t to sender via feedback channel

— ADAPTATION —

- A1. $\pi_s(t+1)(\Gamma) \propto \pi_s(t)(\Gamma) \cdot \exp(\eta_s \cdot \delta_t \cdot \mathcal{R}[\Gamma_t = \Gamma])$ [Sender policy]
- A2. $\theta_r(t+1) \leftarrow \theta_r(t) + \eta_r \cdot \nabla_{\{\theta_r\}} \mathcal{F}_s(t)$ [Receiver params]
- A3. $\mu_s(t+1) \leftarrow \mu_s(t) + \gamma \cdot \delta_t$ [Field permeability]
- A4. Check convergence; $t \leftarrow t + 1$

END WHILE

RETURN $(\hat{C}_s(T), \hat{\mathcal{F}}_s(T), \mu_s(T), \Gamma^*, \hat{r}^*)$

Algorithm 3.1: Complete master algorithm for LPG Semantic Communication (Coupled Adaptive Loop).

3.8.11 Original Contributions of the Master Algorithm

The master algorithm makes four contributions that distinguish it from all prior semantic communication systems including DeepSC and its variants.

1. **Sem-based atomic transmission.** The atomic transmission unit is the quadruple (B, W, Γ, ι) rather than a symbol or a bit sequence. Every other system transmits transformed bits. This system transmits structured meaning.
2. **Analyticity gate.** Step S4 enforces the Cauchy–Riemann semantic regularity conditions before every transmission. This is a mathematical constraint on the structure of meaning being transmitted, with no analogue in any existing communication algorithm.
3. **Semantic conservation enforcement.** Step S6 imposes a physical conservation law at the semantic layer. No existing semantic communication system enforces a conservation principle at the layer of meaning.
4. **Learned semantic permeability.** Step A3 treats μ_s as a learned channel parameter that converges through transmission. The system discovers the semantic structure of the channel through iterative feedback, exactly as an adaptive equalizer discovers the physical channel response.

3.8.12 Recommended Simulation Parameters

Table 3.8.2 provides the recommended parameter ranges for initial simulation. These values are chosen to ensure numerical stability while keeping the adaptive loop tractable on standard GPU hardware.

Parameter	Symbol	Recommended Range	Role
Vocabulary size	$ \mathbf{W} $	500 – 5000	Semantic codebook dimension
Fidelity target	$\mathcal{F}^*$	0.85 – 0.95	Convergence quality threshold
Sender learning rate	η_s	0.01 – 0.1	Anchor policy update step
Receiver learning rate	η_r	10^{-3} – 10^{-2}	Gradient ascent step
Permeability step	γ	0.001 – 0.01	Field parameter adaptation
Noise variance	σ^2	0.01 – 1.0	AWGN channel intensity
Convergence tolerance	ϵ_{conv}	10^{-3} – 10^{-2}	Stopping criterion
Patience window	τ_{conv}	10 – 50 rounds	Consecutive stability required
Max rounds	T_{max}	500 – 5000	Hard stopping limit
Energy drift bound	$\Delta \mathcal{E}$	0.05 – 0.20	Conservation anomaly threshold

Table 3.8.2: Recommended simulation parameters for Algorithm 3.1.

Algorithm 3.1, together with the theoretical structures from which it is derived, constitutes a complete and reproducible specification of LPG Semantic Communication. Any independent investigator equipped with the vocabulary $\mathbf{W}$, the parameter ranges in Table 3.8.2, and a standard Python/Sionna simulation environment can execute this algorithm and recover the semantic

capacity, fidelity, and permeability characterization of any LPG semantic channel. The pseudocode scaffold for this implementation is provided in Appendix I.